LATHE OPERATIONS, THREAD CUTTING, GEAR CUTTING AND TAPER TURNING

Reference: Elements of Workshop Technology, Vol. II - Hajra Choudhury

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1. Lathe Operations

A lathe is a machine tool in which the workpiece rotates about its axis while a cutting tool is fed against it. By controlling the tool path, feed and spindle speed, cylindrical, conical, flat, threaded and formed surfaces can be produced.

Operation	Purpose / description
Facing	Produces a flat surface at the end of the work. The tool is fed radially towards the axis.
Plain / straight turning	Reduces the outside diameter uniformly by feeding the tool parallel to the work axis.
Step turning	Produces two or more cylindrical diameters with shoulders between them.
Taper turning	Produces a conical surface by moving the tool at an angle to the work axis.
Form turning	A form tool having the required profile is fed radially into the work.
Chamfering	Bevels the edge of a cylindrical surface, commonly at 45°, to remove sharp edges.
Grooving / necking	Produces a narrow recess on the external or internal surface.
Parting-off	Cuts the finished part from bar stock by radial feed of a narrow parting tool.
Knurling	Produces a regular rough pattern by plastic deformation using knurling rollers; no metal is removed.
Thread cutting	Produces helical grooves of specified pitch by synchronizing tool feed with spindle rotation.
Drilling	A drill held in the tailstock is fed axially into the rotating work.
Boring	Enlarges and trues an existing hole using a single-point boring tool.
Reaming	Finishes a previously drilled or bored hole to close size and good finish.
Counterboring / countersinking	Produces stepped or conical recesses at the mouth of a hole.
Eccentric turning	Produces cylindrical surfaces whose axes do not coincide with the main work axis.
Polishing / filing	Improves surface finish after machining using abrasive cloth or a file with proper safety precautions.

1.1 Classification of lathe operations

Category	Typical operations
Operations on external surfaces	Turning, step turning, taper turning, form turning, threading, grooving, knurling, chamfering, parting.
Operations on internal surfaces	Drilling, boring, internal threading, reaming, counterboring, countersinking.
Finishing operations	Reaming, polishing, filing and fine turning.

2. Thread / Screw Cutting Mechanism

Thread cutting on a centre lathe is carried out by moving a single-point threading tool longitudinally at a definite rate with respect to the rotating work. The tool must advance by exactly one thread pitch during one revolution of the work.

2.1 Principle

Tool travel per spindle revolution = pitch of thread to be cut

The spindle drives the lead screw through a train of gears. The carriage is connected to the lead screw through half-nuts. When the half-nuts engage the rotating lead screw, the carriage moves longitudinally at a rate fixed by the lead-screw pitch and gear ratio.

2.2 Main elements of the screw-cutting mechanism

Element	Function
Spindle	Rotates the workpiece.
Stud gear / spindle gear	Takes drive from the spindle.
Change gears or quick-change gearbox	Provides the required speed ratio between spindle and lead screw.
Lead screw	Accurately pitched screw that drives the carriage during thread cutting.
Half-nuts	Split nuts in the apron that engage/disengage the lead screw.
Apron and carriage	Carry the cutting tool along the bed.
Thread dial indicator	Assists correct re-engagement of half-nuts for repeated passes, where applicable.

2.3 Change-gear relation

For a simple gear train connecting spindle to lead screw:

Lead-screw revolutions / spindle revolution = Driver teeth / Driven teeth

Required ratio = Lead-screw pitch / Thread pitch to be cut

The exact placement of gears depends on the lathe convention. Compound gearing is used when the required ratio cannot be obtained with one simple pair.

2.4 Procedure for cutting an external thread

1. Turn the work to the correct major diameter and chamfer the starting end.
2. Grind/set the threading tool to the correct included angle and set it at centre height.
3. Set the change gears or quick-change gearbox for the required pitch.
4. Set a low suitable spindle speed.
5. Bring the tool near the work and establish the zero/reference setting.
6. Engage the half-nuts at the correct dial indication and take a light first cut.
7. At the end of the pass, withdraw the tool, reverse/return the carriage as appropriate, and reset for the next cut.
8. Increase depth gradually until the thread reaches the required form and fit.
9. Check pitch and profile with a screw-pitch gauge, mating nut or appropriate measuring method.

2.5 Right-hand and left-hand threads

Thread	Carriage motion during cutting	Remarks
Right-hand thread	Tool generally moves from tailstock end towards headstock for the common setup.	Most commonly used.
Left-hand thread	Tool direction and/or spindle rotation is arranged so helix hand is reversed.	Used where rotation would tend to loosen a right-hand thread.

3. Gear Cutting Mechanism

Gears can be produced by several methods. In workshop practice, gear teeth may be cut on a milling machine using a form-relieved gear cutter and an indexing head. The blank is rotated by a precise fraction after each tooth space is milled. Other production methods include hobbing, shaping and broaching.

3.1 Gear cutting by form milling and indexing

10. Prepare the gear blank to the correct outside diameter and width.
11. Mount the blank on an arbor between centres or in a suitable fixture attached to the dividing head.
12. Select the involute gear cutter corresponding to module/diametral pitch and number of teeth.
13. Set the cutter at the centre height of the gear blank.
14. Set the depth of cut equal to the required tooth depth.
15. Mill the first tooth space.
16. Index the blank by the required fraction of a revolution.
17. Repeat until all tooth spaces are cut.

3.2 Simple indexing relation

For a standard dividing head having a 40:1 worm ratio:

$$\text{Crank turns per tooth} = 40 / Z$$

where Z is the number of teeth required on the gear.

3.3 Examples of indexing

Required teeth Z	Crank movement
20	40/20 = 2 turns per tooth
32	40/32 = 1 1/4 turns per tooth
40	40/40 = 1 turn per tooth
64	40/64 = 5/8 turn per tooth

3.4 Other gear-cutting mechanisms

Method	Basic principle	Typical advantage
Gear hobbing	A rotating hob and gear blank are geared together continuously.	Fast, continuous and suitable for production.
Gear shaping	A pinion- or rack-shaped cutter reciprocates while cutter and blank roll together.	Can cut external and internal gears.
Form milling	Each tooth space is milled separately using a form cutter.	Simple and suitable for small-batch work.
Broaching	A broach containing progressively shaped teeth cuts the complete form.	Very high production for suitable internal forms.

4. Taper Turning

A taper is a uniform increase or decrease in diameter along the length of a workpiece. In taper turning, the cutting tool follows a path inclined to the axis of the work.

$$\text{Taper per unit length} = (D - d) / L$$

$$\tan \alpha = (D - d) / (2L)$$

where D = larger diameter, d = smaller diameter, L = length of taper, and α = half-angle of taper.

4.1 Types of taper

- External taper: conical surface on the outside of a component.
- Internal taper: conical surface inside a hole.
- Short steep taper: comparatively large taper angle over a small length.

- Long gradual taper: small taper angle extending over a large axial length.

5. Taper Turning Methods

Method	Principle	Advantages	Limitations
Form tool method	A broad-nosed tool ground to the required included angle is fed radially into the work.	Simple and quick for short tapers.	High cutting force; unsuitable for long tapers; chatter may occur.
Swivelling compound rest	Compound slide is swivelled to the taper half-angle α and the tool is fed manually along the compound slide.	Good for short/steep tapers; accurate angular setting.	Limited taper length; feed is usually manual.
Tailstock set-over	Tailstock body is offset sideways so the work axis is inclined to the lathe bed; tool moves parallel to bed.	Suitable for long, slight external tapers between centres.	Cannot conveniently produce internal tapers; centre holes can be distorted if offset is excessive.
Taper-turning attachment	Guide bar is set to the taper angle and causes the cross-slide to move laterally while carriage travels longitudinally.	Accurate long external/internal tapers; tailstock remains aligned.	Requires attachment and careful setting.

5.1 Tailstock set-over calculation

If l is the distance between centres and L is the taper length:

$$\textbf{Set-over } S = \frac{l(D - d)}{2L}$$

For a taper extending over the full length between centres, L = l and therefore S = (D - d)/2.

5.2 Compound-rest setting

$$\textbf{Compound-rest swivel angle} = \alpha = \tan^{-1}\left[\frac{(D - d)}{(2L)}\right]$$

6. Feed Mechanisms in a Lathe

The feed mechanism transmits motion from the headstock/spindle to the carriage or cross-slide at a selected rate. During ordinary turning the feed rod is generally used; the lead screw is reserved mainly for accurate thread cutting.

Part	Function
Feed gearbox / quick-change gearbox	Selects different feed rates and thread pitches.
Feed rod	Transmits rotary motion to the apron for ordinary power feed.
Lead screw	Provides precise longitudinal motion during screw cutting.
Apron mechanism	Contains clutches/gears for longitudinal and cross feed.
Rack and pinion	Used for manual carriage traverse.
Half-nuts	Engage carriage with the lead screw during thread cutting.

6.1 Longitudinal and cross feed

- Longitudinal feed: carriage moves parallel to the lathe axis; used for turning.
- Cross feed: cross-slide moves perpendicular to the lathe axis; used for facing and diameter reduction.
- Automatic feed is disengaged before reaching shoulders or limits unless a stop/trip arrangement is provided.

7. Cutting Speed, Feed and Depth of Cut

7.1 Cutting speed

Cutting speed is the linear speed at which the work surface passes the cutting edge. It is usually expressed in m/min.

$$V = \frac{\pi DN}{1000}$$

$$N = 1000V / (\pi D)$$

where V = cutting speed (m/min), D = work diameter (mm), N = spindle speed (rev/min).

7.2 Feed

Feed is the distance the tool advances during one revolution of the work. In turning it is commonly expressed in mm/rev.

$$\text{Feed rate} = fN \ \ (\text{mm/min})$$

where f = feed in mm/rev.

7.3 Depth of cut

$$\text{Depth of cut, } d = (D_1 - D_2) / 2$$

D_1 is the initial diameter and D_2 is the final diameter after one pass. Depth of cut is a radial quantity, not the total reduction in diameter.

7.4 Effect of cutting parameters

Parameter increased	General effect
Cutting speed	Machining time decreases; cutting temperature and tool wear generally increase.
Feed	Material removal rate increases; cutting force and surface roughness generally increase.
Depth of cut	Material removal rate and cutting force increase; machine rigidity becomes more important.

8. Machining Time

Machining time is the time during which the cutting tool is actually removing material. For turning and facing, it is calculated from the cutting travel divided by feed rate.

8.1 Turning time

$$T = L / (fN)$$

where T = machining time in minutes, L = total tool travel in mm including approach/over-travel, f = feed in mm/rev, N = rpm.

8.2 Number of passes

$$\text{Number of passes} = \text{Total radial stock removal} / \text{Depth of cut per pass}$$

The number is rounded upward to the next whole pass if the stock removal is not an exact multiple.

8.3 Facing time

$$T = \text{Radial tool travel} / (fN)$$

For facing from the outer diameter to the centre, radial travel is approximately D/2 plus any allowance.

8.4 Thread-cutting time

For one thread-cutting pass, the time is the thread length divided by the axial feed rate. Since feed per revolution equals thread pitch p:

$$T \text{ per pass} = L / (pN)$$

$$\text{Total thread-cutting time} \approx \text{Number of passes} \times L / (pN)$$

Return time and tool setting time are added separately if required by the problem.

9. Simple Numerical Problems

Problem 1: Spindle speed from cutting speed

A 60 mm diameter bar is turned at a cutting speed of 30 m/min. Find spindle speed.

$$N = 1000V/(\pi D) = (1000 \times 30)/(\pi \times 60)$$

$$N \approx 159.2 \text{ rpm} \approx 159 \text{ rpm}$$

Problem 2: Depth of cut

A bar diameter is reduced from 50 mm to 44 mm in one pass. Find the depth of cut.

$$d = (50 - 44)/2 = 3 \text{ mm}$$

Problem 3: Turning time

A 50 mm diameter bar is turned over a length of 120 mm at 25 m/min and feed 0.25 mm/rev. Assume 5 mm approach allowance. Find machining time.

$$N = (1000 \times 25)/(\pi \times 50) = 159.2 \text{ rpm}$$

$$\text{Total travel } L = 120 + 5 = 125 \text{ mm}$$

$$T = 125/(0.25 \times 159.2) = 3.14 \text{ min}$$

Problem 4: Number of passes and machining time

A 60 mm diameter bar is to be reduced to 52 mm over a length of 150 mm. Depth of cut per pass is 2 mm, cutting speed 35 m/min and feed 0.3 mm/rev. Neglect approach allowance.

$$\text{Total radial stock removal} = (60 - 52)/2 = 4 \text{ mm}$$

$$\text{Number of passes} = 4/2 = 2 \text{ passes}$$

$$N = (1000 \times 35)/(\pi \times 60) = 185.7 \text{ rpm}$$

$$\text{Time per pass} = 150/(0.3 \times 185.7) = 2.69 \text{ min}$$

$$\text{Total machining time} = 2 \times 2.69 = 5.39 \text{ min}$$

Problem 5: Taper half-angle

A taper has larger diameter 50 mm, smaller diameter 40 mm and taper length 100 mm. Find taper per unit length and compound-rest swivel angle.

$$\text{Taper per unit length} = (50 - 40)/100 = 0.10$$

$$\tan \alpha = (50 - 40)/(2 \times 100) = 0.05$$

$$\alpha = \tan^{-1}(0.05) \approx 2.86^\circ$$

Problem 6: Tailstock set-over

A workpiece is held between centres 300 mm apart. A taper from 40 mm to 30 mm is required over 150 mm length. Find tailstock set-over.

$$S = l(D - d)/(2L)$$

$$S = 300(40 - 30)/(2 \times 150) = 10.0 \text{ mm}$$

Problem 7: Thread-cutting change-gear ratio

A lathe has a lead screw of 6 mm pitch and a thread of 2 mm pitch is to be cut. Find the required lead-screw/spindle speed ratio.

$$\text{Required ratio} = \text{Lead-screw pitch} / \text{Thread pitch}$$

$$\textbf{Required ratio} = 6/2 = 3$$

Thus the gear arrangement must provide the corresponding 3:1 ratio according to the machine's driver/driven gear convention.

Problem 8: Simple indexing for gear cutting

Find the crank movement per tooth for cutting a 24-tooth gear using a 40:1 dividing head.

$$\textbf{Crank movement} = 40/Z = 40/24 = 1 + 2/3 \text{ turns}$$

Hence the crank is moved 1 full turn plus 2/3 of a turn for each tooth space. For example, on an 18-hole circle, 2/3 turn corresponds to 12 holes.

Problem 9: Thread-cutting time

A 2 mm pitch thread is cut over 60 mm length at 100 rpm in 8 passes. Neglect return time. Find cutting time.

$$\textbf{Feed during threading} = \text{pitch} = 2 \text{ mm/rev}$$

$$\textbf{Time per pass} = 60/(2 \times 100) = 0.30 \text{ min}$$

$$\textbf{Total time} = 8 \times 0.30 = 2.40 \text{ min}$$

10. Important Points

- In turning, the work rotates while the single-point tool is fed against it.
- Facing feed is radial; turning feed is parallel to the work axis.
- During thread cutting, tool advance per spindle revolution must equal the required thread pitch.
- The lead screw is used for accurate screw cutting; the feed rod is normally used for ordinary power feed.
- For a 40:1 dividing head, simple indexing is given by 40/Z crank turns per tooth.
- Taper per unit length = (D - d)/L and tan α = (D - d)/(2L).
- Form-tool method is suited to short tapers; tailstock set-over is suited to long slight external tapers.
- A taper-turning attachment can produce long accurate tapers without disturbing tailstock alignment.
- Cutting speed V = πDN/1000.
- Depth of cut in turning is half the reduction in diameter.
- Turning time T = L/(fN), with approach/over-travel included in L where specified.

Typical Examination Questions

18. Explain the principal operations performed on a centre lathe.
19. Explain the thread-cutting mechanism of a lathe with the function of lead screw and half-nuts.
20. Derive the change-gear relation used for thread cutting.
21. Explain the method of gear cutting by form milling and simple indexing.
22. What is taper turning? Derive the relation for taper angle.
23. Explain the different methods of taper turning with advantages and limitations.
24. Explain the feed mechanism of a centre lathe.
25. Define cutting speed, feed and depth of cut and write the relevant expressions.
26. Derive the machining-time relation for a turning operation.
27. Solve problems based on spindle speed, taper turning, tailstock set-over, thread cutting, indexing and machining time.

Reference

Hajra Choudhury, S. K. and Hajra Choudhury, A. K., Elements of Workshop Technology, Vol. II.